HOTA Metric — Compact Algorithm + Variable Reference

Compact Paper-Style Algorithm

Per Video / Sequence

Input

$$\{\mathcal{GT}_t, \mathcal{P}red_t, S^t\}_{t=1}^{T}, \qquad \mathcal{A} = \{0.05, 0.10, \ldots, 0.95\}$$

Total G groundtruth tracks and P prediction tracks

1. Similarity normalization

For frame t, ground-truth detection i, and prediction j:

$$R_{ij}^t = \frac{S_{ij}^t}{\sum_{i=1}^{N_t} S_{ij}^t + \sum_{j=1}^{M_t} S_{ij}^t - S_{ij}^t}$$

Shape of R^t : $N_t \times M_t$

2. Accumulate global identity co-occurrence

For every GT identity g and predicted identity p:

$$C_{gp} = \sum_{t,i,j: ID_G(i)=g,\ ID_P(j)=p} R_{ij}^t.$$

Also count detections belonging to each identity over the whole video:

$$N_g^G = \sum_t \#\{i : ID_G(i) = g\},$$

$$N_p^P = \sum_t \#\{j : ID_P(j) = p\}.$$

Shape of C : $G \times P$

3. Global identity alignment

$$\boxed{A_{gp} = \frac{C_{gp}}{N_g^G + N_p^P - C_{gp}}}$$

Shape of A : $G \times P$

This score measures how strongly GT identity g aligns with predicted identity p over the complete video.

4. Frame-level matching score

For every pair (i, j) in frame t:

Per-frame quantities

Variable	Meaning	Shape
$\mathcal{GT}_t$	GT detections in frame t	set/list of size G_t
$\mathcal{P}red_t$	Predicted detections in frame t	set/list of size P_t
S^t	Localization similarity matrix	$G_t \times P_t$
T	Number of frames in the video	scalar
N_t	Number of GT detections in frame t	scalar
M_t	Number of predicted detections in frame t	scalar
S_{ij}^t	Similarity between GT detection i and prediction j	scalar
$ID_G(i)$	GT track identity of detection i	scalar ID
$ID_P(j)$	Predicted track identity of detection j	scalar ID

Whole-video identity quantities

Variable	Meaning	Shape
G	Number of unique GT identities	scalar
P	Number of unique predicted identities	scalar
N^G	Total detection count for every GT identity	G
N_g^G	Number of GT detections belonging to identity g across the video	scalar
N^P	Total detection count for every predicted identity	P
N_p^P	Number of predictions belonging to identity p across the video	scalar
A	Global identity alignment matrix	$G \times P$

$$\boxed{W_{ij}^t = A_{ID_G(i),\, ID_P(j)} \cdot S_{ij}^t}$$

5. Optimal one-to-one matching

Apply Hungarian matching:

$$\boxed{\Pi_t = \arg\max_{\Pi} \sum_{(i,j)\in\Pi} W_{ij}^t}$$

subject to every GT detection and prediction being used at most once.

6. Threshold accepted matches

For every threshold $\alpha \in \mathcal{A}$:

$$\boxed{\Pi_t^\alpha = \{(i,j) \in \Pi_t : S_{ij}^t \geq \alpha\}}$$

Only Hungarian matches satisfying the original localization threshold are retained.

7. Detection counts

$$\boxed{TP_\alpha = \sum_t |\Pi_t^\alpha|}$$

$$\boxed{FN_\alpha = \sum_t (N_t - |\Pi_t^\alpha|)}$$

$$\boxed{FP_\alpha = \sum_t (M_t - |\Pi_t^\alpha|)}$$

8. Identity-pair match counts

For every GT identity g and predicted identity p:

$$\boxed{M_{gp}^\alpha = \sum_{t=1}^{T} \#\{(i,j) \in \Pi_t^\alpha : ID_G(i) = g,\ ID_P(j) = p\}}$$

Equivalently,

$$M_{gp}^\alpha = \text{number of accepted detection matches across the video}$$

whose GT detection belongs to identity g and whose prediction belongs to identity p.

Thus,

$$M_{gp}^\alpha = 3$$

means GT track g and predicted track p were accepted as a matched pair in **three frames** at threshold α.

Variable	Meaning	Shape
A_{gp}	Alignment score between GT identity g and predicted identity p	scalar

Matching quantities

Variable	Meaning	Shape
W^t	HOTA matching-score matrix	$N_t \times M_t$
W_{ij}^t	Matching score $A_{gp}S_{ij}^t$	scalar
Π_t	Hungarian one-to-one matches in frame t	set of pairs
Π_t^α	Hungarian matches accepted after threshold α	set of pairs
α	Localization threshold	scalar
$\mathcal{A}$	Set of evaluation thresholds	19 scalars

Identity-pair match counts

Variable	Meaning	Shape
M^α	Accepted identity-pair match-count matrix	$G \times P$
M_{gp}^α	Number of accepted detection matches between GT identity g and prediction identity p across all frames	scalar

Intuition

If

$$M_{gp}^\alpha = 5,$$

then five accepted detection pairs across the video connected GT identity g with predicted identity p at threshold α.

This quantity is therefore the bridge between **frame-level detection matching** and **track-level association quality**.

Per-threshold detection counts

Variable	Meaning	Shape
TP_α	Total accepted matched detections	scalar
FN_α	Total unmatched GT detections	scalar

9. Association Accuracy

For an identity pair (g, p):

$$AssIoU_{gp}^{\alpha} = \frac{M_{gp}^{\alpha}}{N_g^G + N_p^P - M_{gp}^{\alpha}}$$

Then

$$\boxed{AssA_{\alpha} = \frac{\sum_{g,p} M_{gp}^{\alpha}\, AssIoU_{gp}^{\alpha}}{\max(1, TP_{\alpha})}}$$

or directly,

$$AssA_{\alpha} = \frac{\sum_{g,p} M_{gp}^{\alpha} \frac{M_{gp}^{\alpha}}{N_g^G + N_p^P - M_{gp}^{\alpha}}}{\max(1, TP_{\alpha})}.$$

10. Localization Accuracy

Accumulate localization similarity of accepted matches:

$$L_{\alpha} = \sum_t \sum_{(i,j)\in\Pi_t^{\alpha}} S_{ij}^t.$$

Then

$$\boxed{LocA_{\alpha} = \frac{\max(10^{-10}, L_{\alpha})}{\max(10^{-10}, TP_{\alpha})}}$$

If TP=0, then LocA = 0

11. Association Recall and Precision

$$\boxed{AssRe_{\alpha} = \frac{\sum_{g,p} M_{gp}^{\alpha} \frac{M_{gp}^{\alpha}}{N_g^G}}{\max(1, TP_{\alpha})}}$$

$$\boxed{AssPr_{\alpha} = \frac{\sum_{g,p} M_{gp}^{\alpha} \frac{M_{gp}^{\alpha}}{N_p^P}}{\max(1, TP_{\alpha})}}$$

12. Detection metrics

$$\boxed{DetRe_{\alpha} = \frac{TP_{\alpha}}{TP_{\alpha} + FN_{\alpha}}}$$

$$\boxed{DetPr_{\alpha} = \frac{TP_{\alpha}}{TP_{\alpha} + FP_{\alpha}}}$$

$$\boxed{DetA_{\alpha} = \frac{TP_{\alpha}}{TP_{\alpha} + FN_{\alpha} + FP_{\alpha}}}$$

Variable	Meaning	Shape
FP_{α}	Total unmatched predicted detections	scalar
L_{α}	Sum of localization similarities of accepted matches	scalar

Per-threshold metrics

Variable	Meaning	Shape
$AssIoU_{gp}^{\alpha}$	Association IoU/Jaccard for identity pair (g, p)	scalar
$AssA_{\alpha}$	Association Accuracy	scalar
$AssRe_{\alpha}$	Association Recall	scalar
$AssPr_{\alpha}$	Association Precision	scalar
$LocA_{\alpha}$	Localization Accuracy	scalar
$DetRe_{\alpha}$	Detection Recall	scalar
$DetPr_{\alpha}$	Detection Precision	scalar
$DetA_{\alpha}$	Detection Accuracy	scalar
$HOTA_{\alpha}$	HOTA at threshold α	scalar

Multi-video quantities

Variable	Meaning	Shape
K	Number of videos/sequences	scalar
k	Video index	scalar
$TP_{\alpha}^{(k)}$	TP count from video k	scalar
$FN_{\alpha}^{(k)}$	FN count from video k	scalar
$FP_{\alpha}^{(k)}$	FP count from video k	scalar
$AssA_{\alpha}^{(k)}$	Association Accuracy of video k	scalar
$AssRe_{\alpha}^{(k)}$	Association Recall of video k	scalar
$AssPr_{\alpha}^{(k)}$	Association Precision of video k	scalar
$LocA_{\alpha}^{(k)}$	Localization Accuracy of video k	scalar

13. HOTA

$$\boxed{HOTA_\alpha = \sqrt{DetA_\alpha \cdot AssA_\alpha}}$$

Combine Multiple Videos / Sequences

Let the dataset contain

$$V_1, V_2, \ldots, V_K.$$

1. Sum detection counts

$$TP_\alpha = \sum_{k=1}^{K} TP_\alpha^{(k)}$$

$$FN_\alpha = \sum_{k=1}^{K} FN_\alpha^{(k)}$$

$$FP_\alpha = \sum_{k=1}^{K} FP_\alpha^{(k)}.$$

2. TP-weighted association metrics

For

$$X \in \{AssA, AssRe, AssPr\},$$

$$\boxed{X_\alpha = \frac{\sum_{k=1}^{K} TP_\alpha^{(k)} X_\alpha^{(k)}}{TP_\alpha}}$$

3. TP-weighted localization accuracy

$$\boxed{LocA_\alpha = \frac{\sum_{k=1}^{K} TP_\alpha^{(k)} LocA_\alpha^{(k)}}{TP_\alpha}}$$

4. Recompute dataset detection metrics

$$DetRe_\alpha = \frac{TP_\alpha}{TP_\alpha + FN_\alpha}$$

$$DetPr_\alpha = \frac{TP_\alpha}{TP_\alpha + FP_\alpha}$$

$$DetA_\alpha = \frac{TP_\alpha}{TP_\alpha + FN_\alpha + FP_\alpha}$$

5. Recompute dataset HOTA

$$\boxed{HOTA_\alpha = \sqrt{DetA_\alpha \cdot AssA_\alpha}}$$

Also:

$$OWTA_\alpha = \sqrt{DetRe_\alpha \cdot AssA_\alpha}.$$

OWTA score focuses only on matched detections by not inluding FP's

Final reported HOTA

$$\boxed{HOTA = \frac{1}{19} \sum_{\alpha=0.05}^{0.95} HOTA_\alpha}$$

where α increases in steps of 0.05.